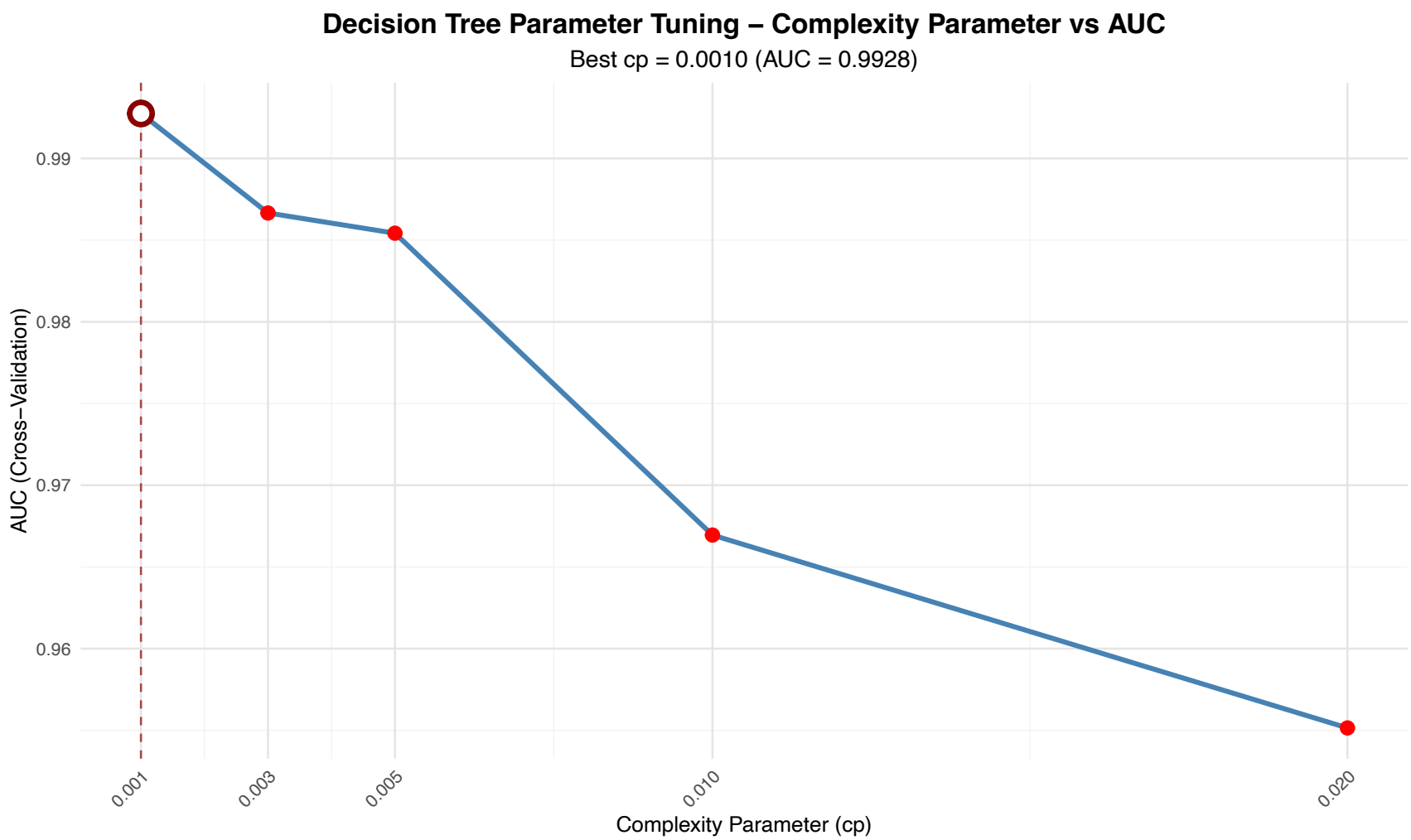
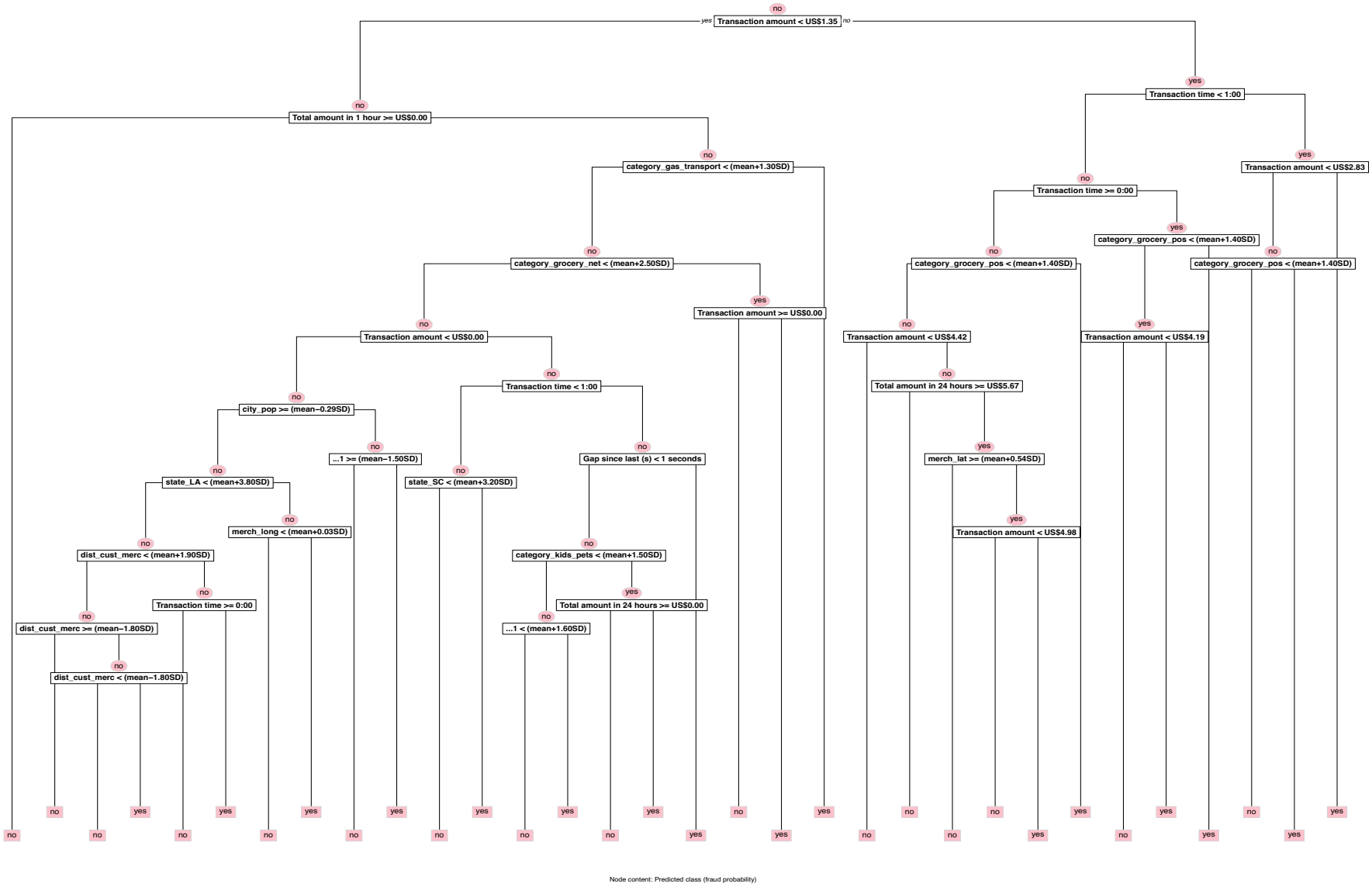
Appendix 1

(Figure A.1)

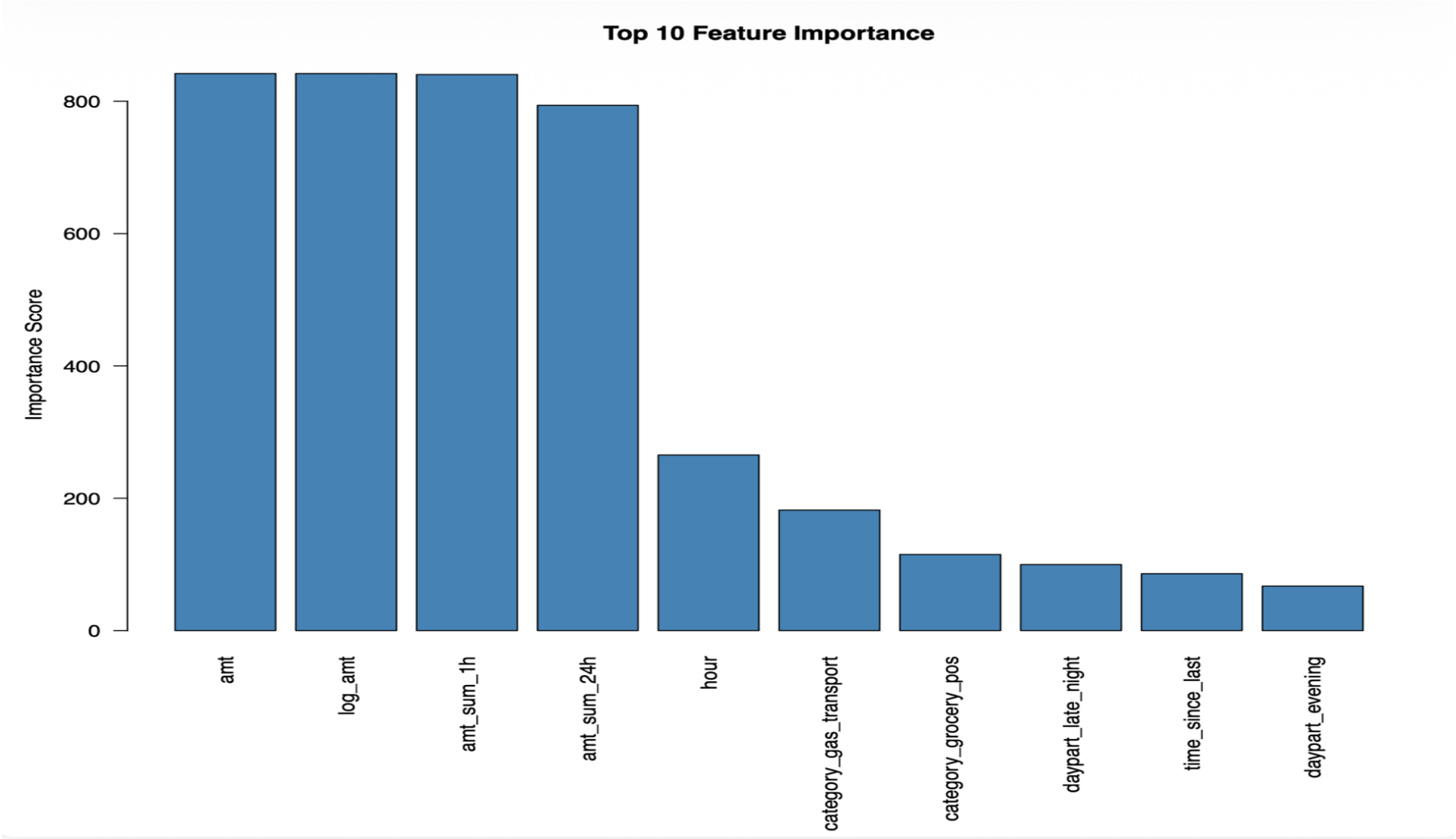


(Figure A.2)

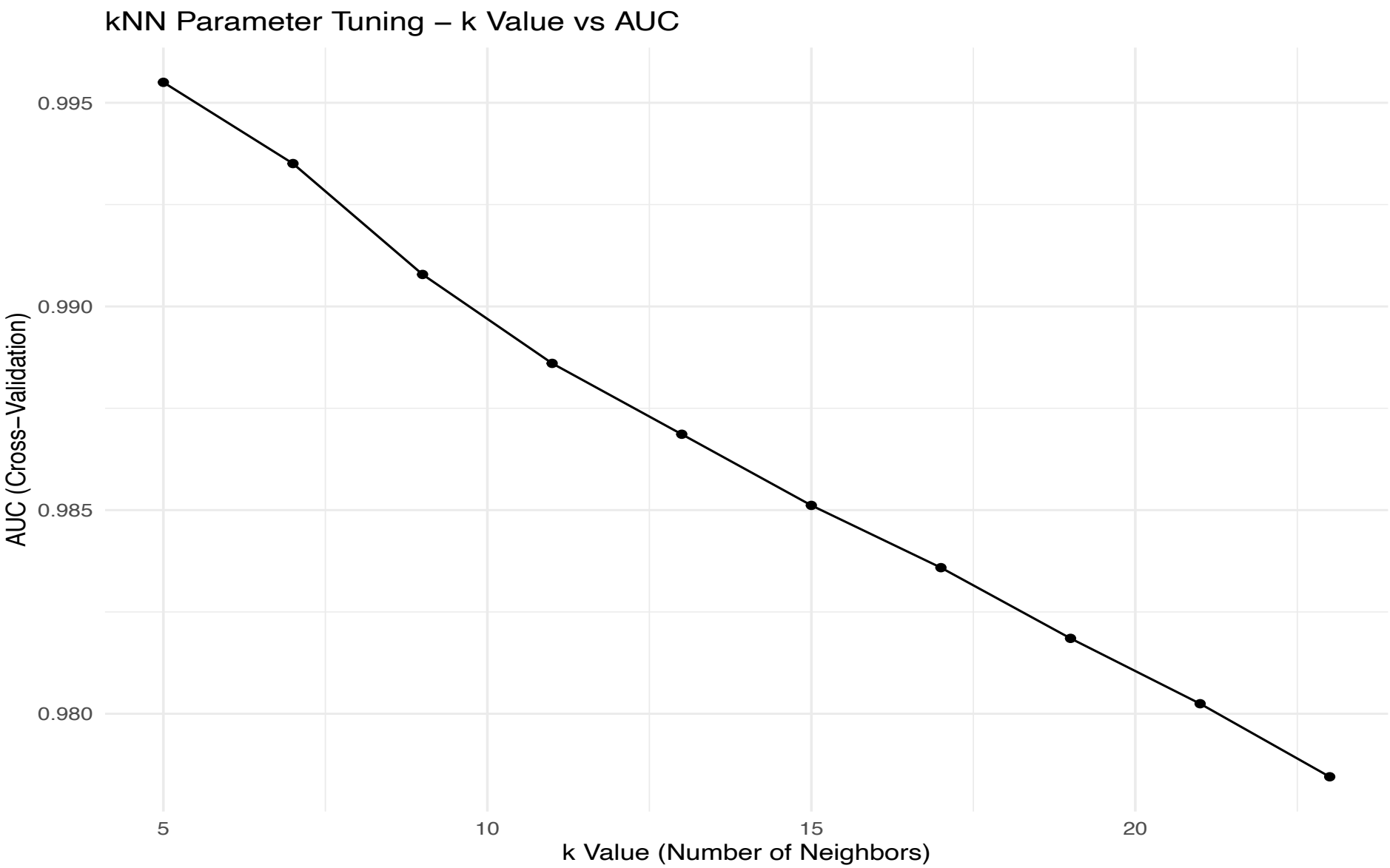


Node content: Predicted class (fraud probability)

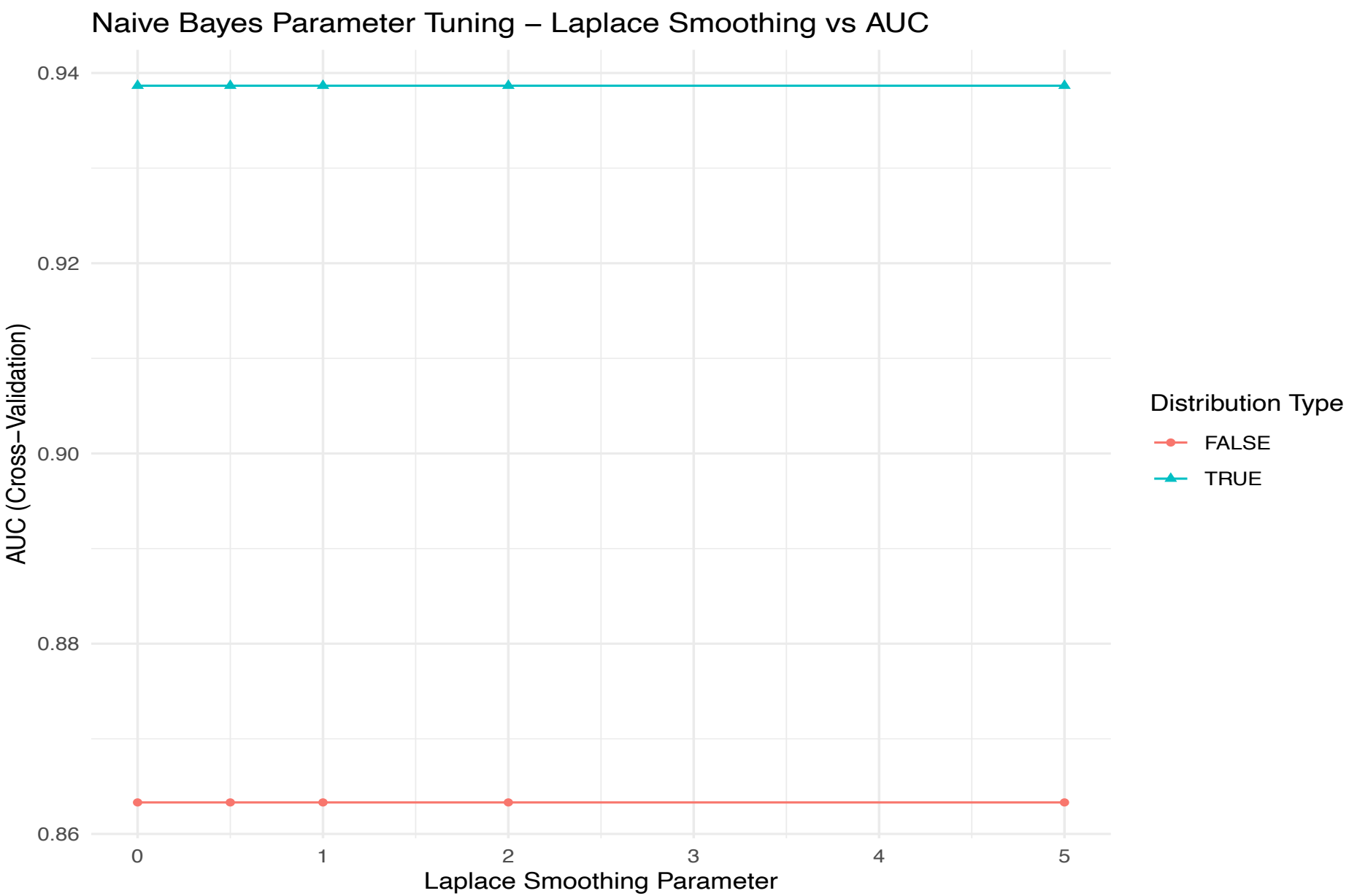
(Figure A.3)


Top 10 Feature Importance
Importance Score
800
600
400
200
0
amt
log_amt
amt_sum_1h
amt_sum_24h
hour
category_gas_transport
category_grocery_pos
daypart_late_night
time_since_last
daypart_evening

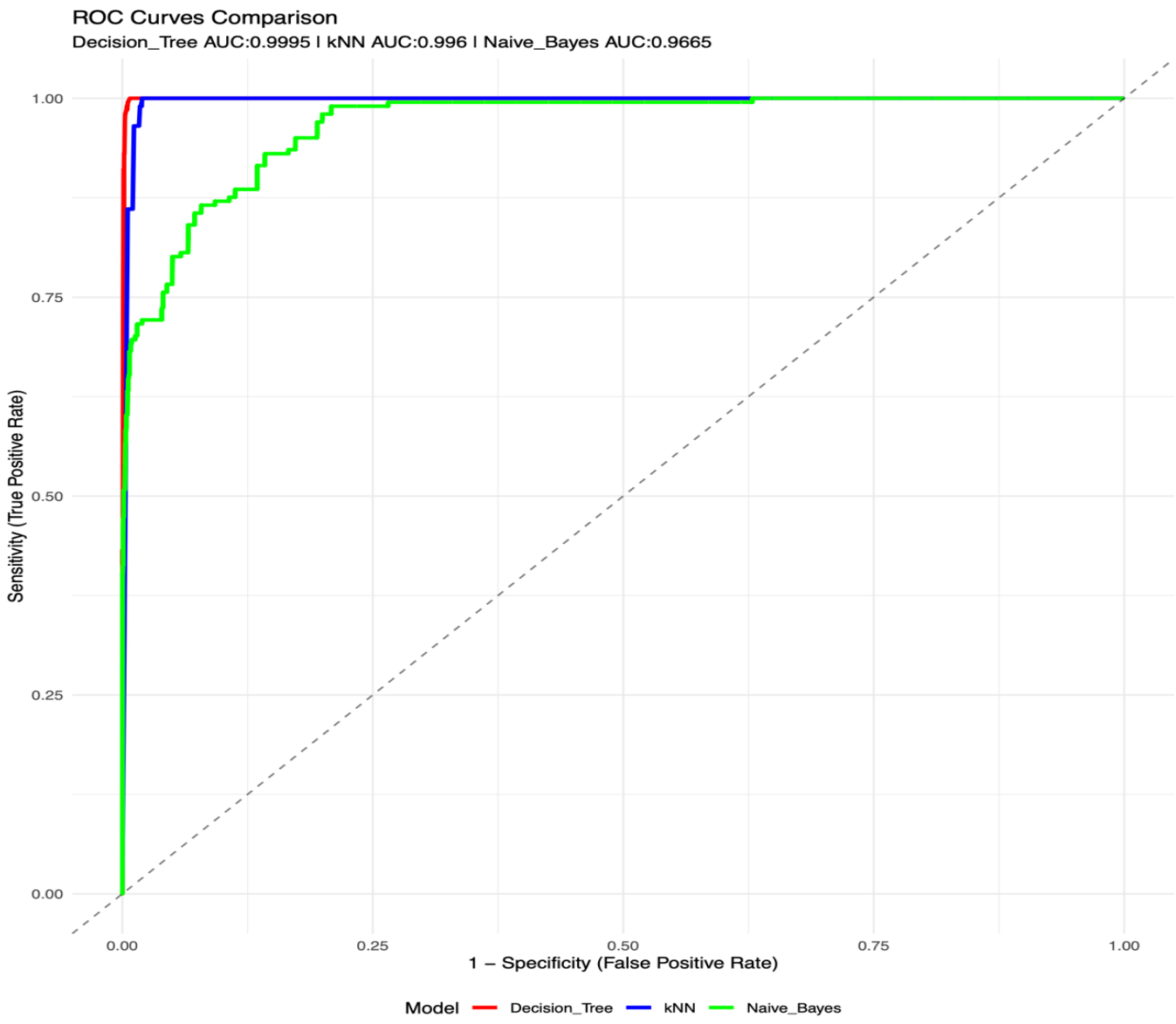
(Figure A.4)


kNN Parameter Tuning – k Value vs AUC
AUC (Cross–Validation)
0.995
0.990
0.985
0.980
5
10
15
20
k Value (Number of Neighbors)

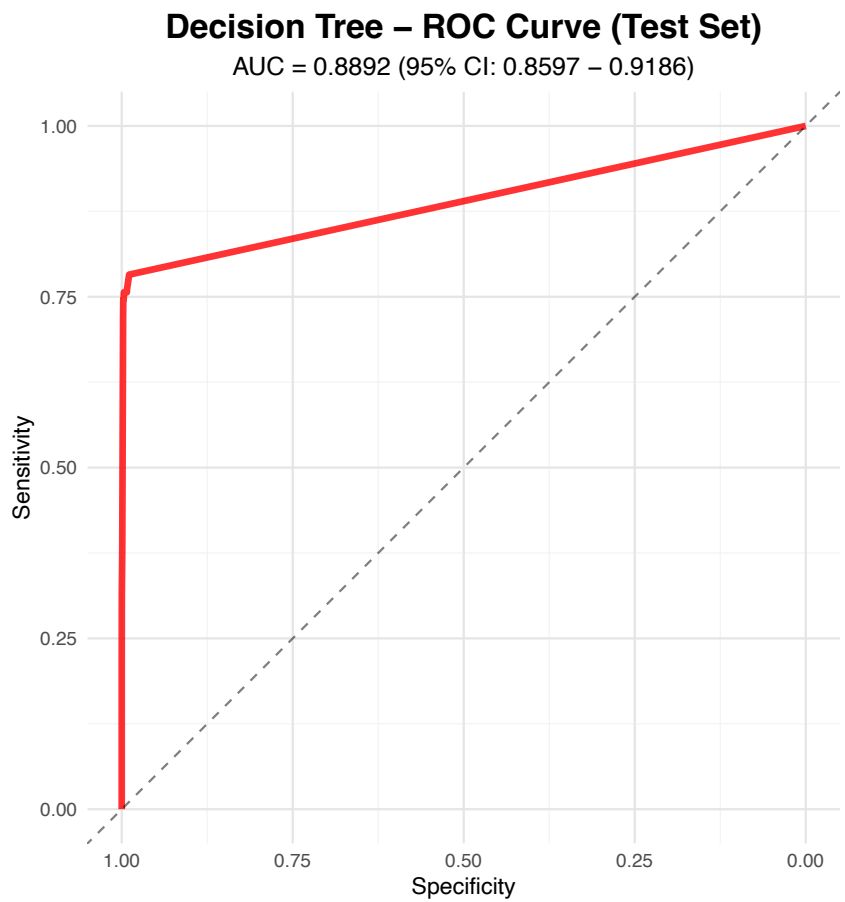
(Figure A.5)

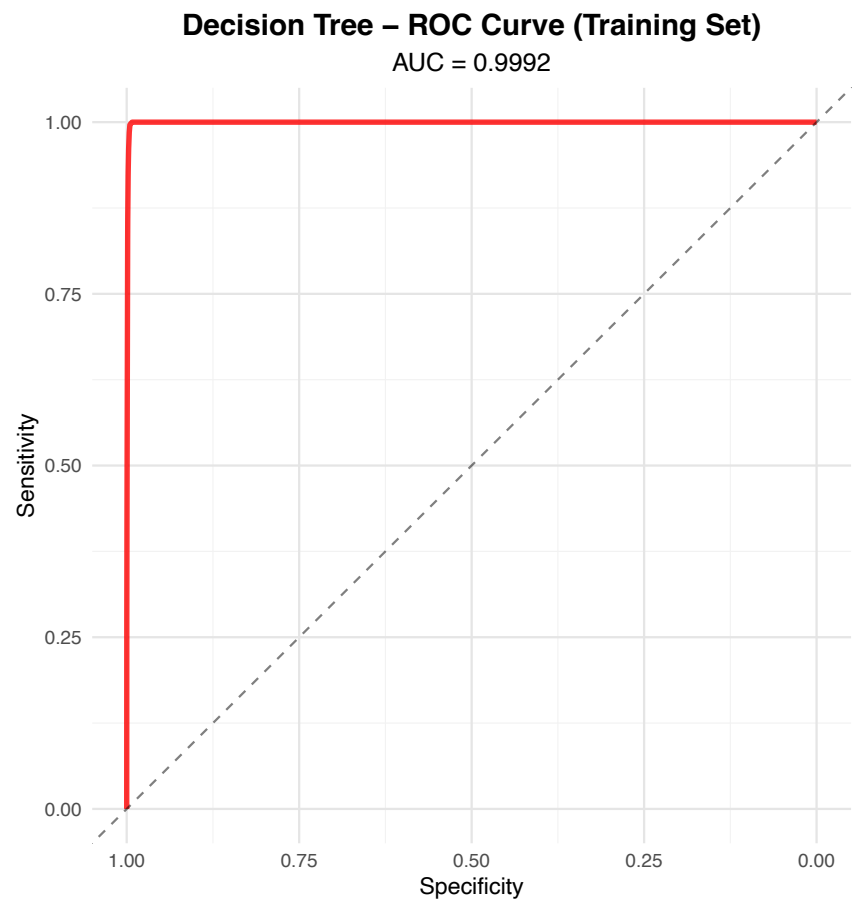

Naive Bayes Parameter Tuning – Laplace Smoothing vs AUC
0.94
0.92
0.90
0.88
0.86
AUC (Cross–Validation)
0
1
2
3
4
5
Laplace Smoothing Parameter
Distribution Type
FALSE
TRUE

(Figure A.6)


ROC Curves Comparison
Decision_Tree AUC:0.9995 I kNN AUC:0.996 I Naive_Bayes AUC:0.9665
Sensitivity (True Positive Rate)
1.00
0.75
0.50
0.25
0.00
0.00
0.25
0.50
0.75
1.00
1 – Specificity (False Positive Rate)
Model
Decision_Tree
kNN
Naive_Bayes

(Figure A.7)


Decision Tree – ROC Curve (Test Set)
AUC = 0.8892 (95% CI: 0.8597 – 0.9186)
Sensitivity
Specificity
1.00
0.75
0.50
0.25
0.00


Decision Tree – ROC Curve (Training Set)
AUC = 0.9992
Sensitivity
Specificity
1.00
0.75
0.50
0.25
0.00